

RESULTS

CALCULATING METRICS

M1 – M2: Embedding Time and Index Size

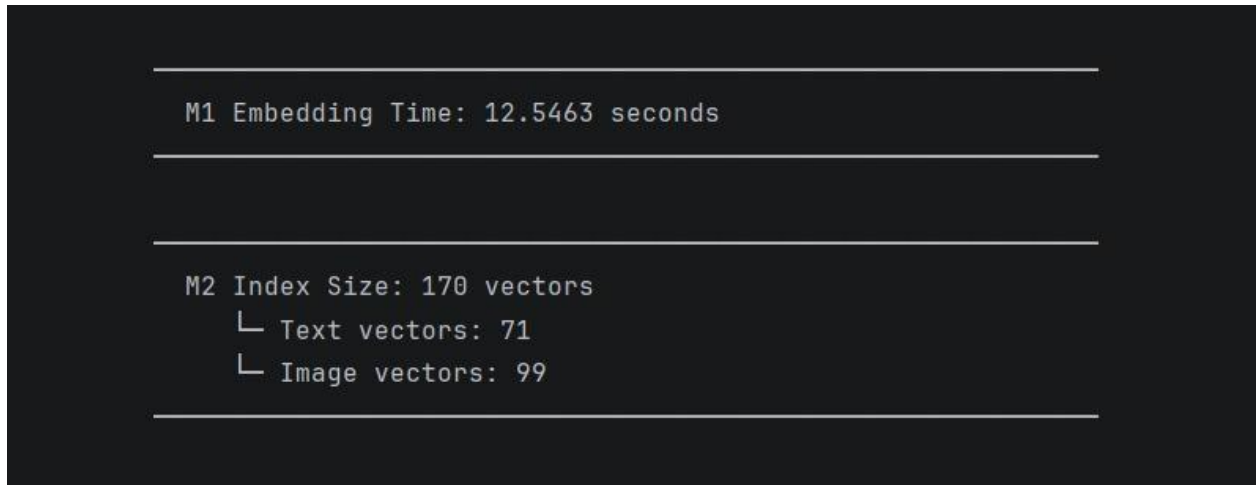


Figure 1: Embedding Time and Index Size Output

M3 – M5: Retrieval Metrics

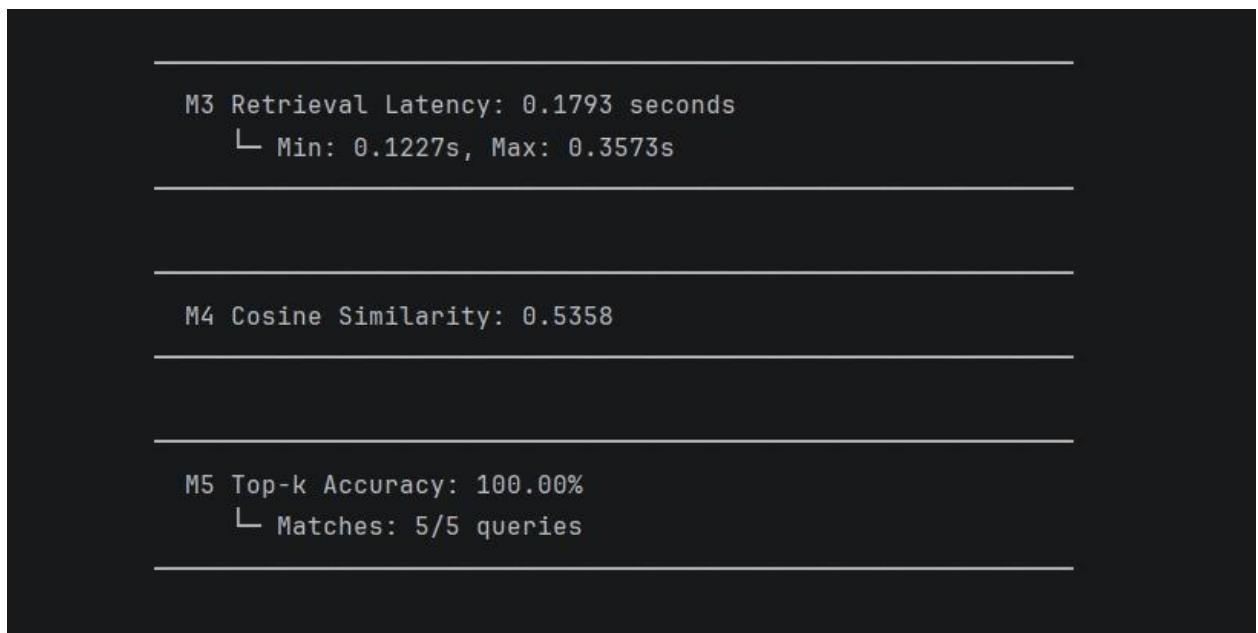


Figure 2: Retrieval Quality Metrics Output

M6 – M8 & M15: Generation Metrics

M6 ROUGE-1: 0.4861 Evaluated: 5, Skipped: 0
M7 ROUGE-2: 0.2382 Evaluated: 5, Skipped: 0
M8 ROUGE-L: 0.2541 Evaluated: 5, Skipped: 0
M15 Context Coverage: 33.60% Evaluated: 5, Skipped: 0

Figure 3: Generation Quality Metrics Output

M9 – M19: Context & System Performance Metrics

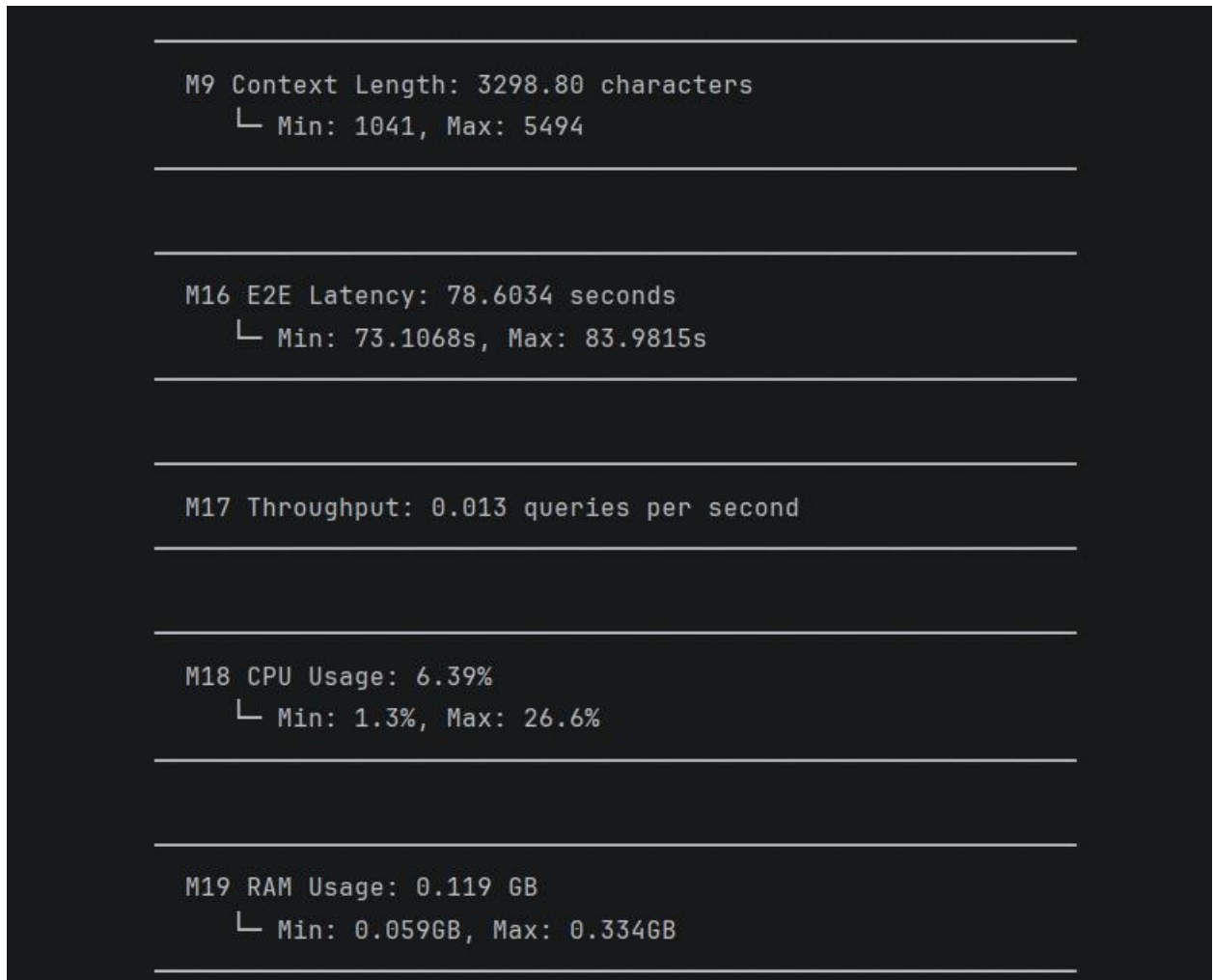


Figure 4: System Performance Metrics Output

FINAL METRICS SUMMARY

Embedding Statistics

EMBEDDING STATISTICS	
Text Embeddings (BGE-large-en-v1.5)	Dim: 1024
Documents processed:	71
Total time:	4.6819 seconds
Avg time/document:	0.0659 seconds
Image Embeddings (ViT-L-14)	Dim: 768
Images processed:	99
Total time:	7.8644 seconds
Avg time/image:	0.0794 seconds
TOTAL EMBEDDING TIME (M1)	12.5463 seconds

Figure 5: Embedding Statistics Summary

Final Metrics Table

METRIC	VALUE
EFFICIENCY	
M1 Embedding Time	12.5463 s
M2 Index Size	170
RETRIEVAL QUALITY	
M3 Retrieval Latency	0.1793 s
M4 Cosine Similarity	0.5358
M5 Top-k Accuracy	100.00 %
GENERATION QUALITY	
M6 ROUGE-1	0.4861
M7 ROUGE-2	0.2382
M8 ROUGE-L	0.2541
M15 Context Coverage	33.60 %
CONTEXT & PERFORMANCE	
M9 Context Length	3298.80 ch
M16 E2E Latency	78.6034 s
M17 Throughput	0.013 q/s
M18 CPU Usage	6.39 %
M19 RAM Usage	0.119 GB
GPU Usage	71.50 %

Figure 6: Final Metrics Summary Table